Unit Test Report

Project: File Compression Tool (Basic ZIP Implementation): Team 14
Date of execution: 05-Nov-2026
Code version: `main` @ `cae72ce` + Sprint-3 tests (`tests/test_robustness.cpp`)
Related: Test Case Document v2.0 (Section 4), STP v1.1 Section 5 (Unit testing)

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1. Summary

Test group	File	Owner	Tests	Passed	Failed
Huffman tree, codes, encoding	`tests/test_huffman.cpp`	Pavan	7	7	0
Input validation and frequencies	`tests/test_input.cpp`	Pranavi	4	4	0
Codec round trips and rejection	`tests/test_codec.cpp`	Rachit	7	7	0
Robustness and error paths	`tests/test_robustness.cpp`	Sprint 3	18	18	0
Total			**36**	**36**	**0**

Result: PASS (36/36). The runner exits with code 0.

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2. Test environment

Item	Value
Test runner	Built-in runner, no external libraries (`tests/test_main.cpp` , SAD 3.6)
Command	`make test` (builds `run_tests` from `src/*.cpp` except `main.cpp` , plus `tests/*.cpp`)
Machine 1	Ubuntu 24.04.5 LTS, x86-64, GCC 13.3.0 (`-std=c++17 -Wall -Wextra -O2`)
Machine 2	Same machine, Clang 18.1.3 with libc++ (the standard library used on macOS)
Memory/UB check	GCC with `-fsanitize=address,undefined`
CI	GitHub Actions on `ubuntu-latest` , `windows-latest` , `macos-latest` (`.github/workflows/build.yml`), every push and pull request

Build	Warnings	Result
GCC 13.3.0, `-Wall -Wextra -Wpedantic`	0	36/36 PASS
Clang 18.1.3 + libc++	0	36/36 PASS
GCC + AddressSanitizer + UBSan	0 sanitizer errors	36/36 PASS

3. Detailed results

3.1 Huffman module (`src/huffman.cpp`): REQ-3

ID	Test function	What is checked	Result
UT-HUF-01	`test_empty_frequencies`	All frequencies zero → `build_tree` returns no tree	PASS
UT-HUF-02	`test_single_symbol`	One symbol → the tree is a single leaf, code `"0"`	PASS
UT-HUF-03	`test_basic_huffman_codes`	A=5, B=2, C=1 → every symbol gets a code	PASS
UT-HUF-04	`test_deterministic_tie_breaking`	Four equal frequencies → exactly A=`00`, B=`01`, C=`10`, D=`11` on every compiler (D-09)	PASS
UT-HUF-05	`test_binary_symbols`	Byte values 0 and 255 encode; packed size = ⌈bits/8⌉	PASS
UT-HUF-06	`test_encoded_size_consistency`	Packed byte count matches the bit count	PASS
UT-HUF-07	`test_missing_code_throws`	Encoding a byte that has no code throws `invalid_argument`	PASS

3.2 Input and frequency module (`src/file_input.cpp`, `src/frequency.cpp`): REQ-1, REQ-2

ID	Test function	Test case	What is checked	Result
UT-IN-01	`tc_comp_01_valid_path`	TC-COMP-01	An existing file is accepted by `validate_input_file`	PASS
UT-IN-02	`tc_err_01_invalid_path`	TC-ERR-01	A missing file and an empty path are rejected with a "not found" message	PASS
UT-IN-03	`tc_comp_02_frequencies`	TC-COMP-02	`aaabbc\n` → a=3, b=2, c=1, `\n`=1, z=0; file read in binary mode (7 bytes)	PASS
UT-IN-04	`tc_comp_02_binary_and_empty`	TC-COMP-02	All 256 byte values counted once; empty input gives all zeros	PASS

3.3 Codec (`src/codec.cpp`): REQ-4 to REQ-7

ID	Test	What is checked	Result
UT-COD-01	Round trip: text	`compress_file` → `decompress_file` → bytes identical	PASS
UT-COD-02	Round trip: empty	0-byte file restores to 0 bytes	PASS
UT-COD-03	Round trip: binary	4 × all 256 byte values restore exactly	PASS
UT-COD-04	Round trip: single byte	1000 × `A` restores exactly	PASS

ID	Test	What is checked	Result
UT-COD-05	Rejects corrupted metadata	Original-size field zeroed → rejected, no output file	PASS
UT-COD-06	Rejects trailing data	One extra byte after the payload → rejected (D-17)	PASS
UT-COD-07	Rejects truncated single-symbol archive	Header only, payload removed → rejected (D-17)	PASS

3.4 Robustness and error paths (Sprint 3): SRS 6.2, SAD 4.4, STP 5.1

Each test damages an archive or blocks an output and checks for **the right exit code and no output file or `.tmp` file left behind**.

ID	What is checked	Expected	Result
UT-ROB-01	Archive file does not exist	`EXIT_FILE_ERROR` (2)	PASS
UT-ROB-02	Header cut inside the original-size field (4 bytes)	Rejected (3)	PASS
UT-ROB-03	Header cut inside the frequency table (500 bytes)	Rejected (3)	PASS
UT-ROB-04	Header cut inside the bit-count field (2060 bytes)	Rejected (3)	PASS
UT-ROB-05	Frequencies add up past 2^{64} (overflow)	Rejected (3)	PASS
UT-ROB-06	Empty archive claiming a non-zero bit count	Rejected (3)	PASS
UT-ROB-07	Non-empty archive claiming zero bits	Rejected (3)	PASS
UT-ROB-08	Single-symbol archive with the wrong bit count	Rejected (3)	PASS
UT-ROB-09	Single-symbol payload containing a `1` bit	Rejected (3)	PASS
UT-ROB-10	Bit count one short (stream ends inside a code)	Rejected (3)	PASS
UT-ROB-11	Extra encoded symbols beyond the original size	Rejected (3)	PASS
UT-ROB-12	Payload missing its last byte	`EXIT_FILE_ERROR` (2)	PASS
UT-ROB-13	Bit count larger than the payload	Rejected (non-zero)	PASS
UT-ROB-14	Compress a file that cannot be read	`EXIT_FILE_ERROR` (2), no output	PASS
UT-ROB-15	Compress into a folder that does not exist	`EXIT_FILE_ERROR` (2)	PASS
UT-ROB-16	Decompress into a folder that does not exist	`EXIT_FILE_ERROR` (2)	PASS
UT-ROB-17	`build_tree` with frequencies that overflow when added	Throws `overflow_error`	PASS
UT-ROB-18	`encode` with a code containing a character other than 0/1	Throws `invalid_argument`	PASS

4. Defects found by unit and system testing during the project

Defect	Found by	Fixed in	Regression test
Huffman codes differed between GCC and Clang/libc++ (cross-OS decoding failed)	Cross-compiler round trip, 21-Oct	PR #24	UT-HUF-04, TC-NF-03
Output could overwrite the input file	Code review / CLI testing, 21-Oct	PR #23	TC-ERR-05
Trailing data and empty single-symbol payloads accepted after the streaming rewrite	Corruption tests, 28-Oct	PR #33	UT-COD-06, UT-COD-07, TC-ERR-03, TC-ERR-04

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5. How to reproduce

```
make clean
make test
```

Expected last line: `All tests passed`.

Appendix: console output (05-Nov-2026, GCC 13.3.0)

Compression/decompression progress messages are omitted; all `[PASS]` lines are shown.

```
Running project tests...
[PASS] Huffman tests                                        (UT-HUF-01..07)
[PASS] Input and frequency tests (TC-COMP-01, TC-ERR-01, TC-COMP-02)
(UT-IN-01..04)
[PASS] Codec round trip: text
[PASS] Codec round trip: empty
[PASS] Codec round trip: binary
[PASS] Codec round trip: single_byte
[PASS] Codec rejects corrupted metadata
[PASS] Codec rejects trailing data
[PASS] Codec rejects truncated single-symbol archive
[PASS] UT-ROB-01 missing archive file -> EXIT_FILE_ERROR
[PASS] UT-ROB-02 header cut inside original-size field is rejected
[PASS] UT-ROB-03 header cut inside frequency table is rejected
[PASS] UT-ROB-04 header cut inside bit-count field is rejected
[PASS] UT-ROB-05 frequency total overflow is rejected
[PASS] UT-ROB-06 empty archive with non-zero bit count is rejected
[PASS] UT-ROB-07 non-empty archive with zero bit count is rejected
[PASS] UT-ROB-08 single-symbol archive with wrong bit count is rejected
[PASS] UT-ROB-09 single-symbol payload containing a 1-bit is rejected
[PASS] UT-ROB-10 bitstream ending in the middle of a code is rejected
[PASS] UT-ROB-11 extra encoded symbols beyond the original size are rejected
[PASS] UT-ROB-12 payload missing its last byte is rejected (truncated bitstream)
[PASS] UT-ROB-13 bit count larger than the payload is rejected
[PASS] UT-ROB-14 compress with unreadable input -> EXIT_FILE_ERROR, no output
[PASS] UT-ROB-15 compress into a missing folder -> EXIT_FILE_ERROR
[PASS] UT-ROB-16 decompress into a missing folder -> EXIT_FILE_ERROR
[PASS] UT-ROB-17 build_tree detects frequency overflow
[PASS] UT-ROB-18 encode rejects a code containing characters other than 0/1
All tests passed
```